

AI & Machine Learning for Civil Engineers

From engineering data to prediction, explainability and uncertainty, in eight hands-on sessions.

08 Live sessions**90** min Per session**04** weeks Course duration**Runs 100% on Google Colab**

Nothing to install, works even on a Chromebook

**Built for beginners**

No prior Python or machine learning required

**Live code-along on real data**

Concrete, geotechnical, water and transport datasets

**Notebooks, assignments & solutions**

Plus a certificate and session recordings

INSTRUCTOR**Dr. Abdul Waris**

Research Associate

Universität Siegen (University of Siegen)

Former Prime Minister's Research Fellow

Ph.D., IIT Hyderabad



Research Excellence Award 2026, IIT Hyderabad



Invited talks at Oregon State University (USA), Norway & Spain

Course outline

Problem → Data → Model → Validate → Interpret → Decide

01 Python for civil engineers

AI/ML basics, Colab, NumPy & pandas on concrete data

02 From raw data to ML-ready

Explore, clean & split data; catch outliers and leakage

03 Your first prediction model

Linear regression, error metrics & cross-validation

04 Trees & ensemble models

Decision trees, Random Forest & boosting for nonlinear data

05 Classification for decisions

Logistic regression, confusion matrix, precision & recall

06 Artificial neural networks

Architecture, activations, loss, epochs & overfitting

07 Explainable AI & engineering judgement

Feature importance & SHAP on a geotechnical case study

08 Uncertainty & reliability analysis

Monte Carlo with ML surrogates to estimate failure probability

WHO SHOULD JOIN

B.Tech & M.Tech students / PhD scholars / Faculty
Practising engineers / Complete beginners to AI & ML

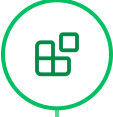
REGISTER NOW · LIMITED SEATS

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 Starts Saturday, 17 October 2026

Most ML courses start with algorithms. This one starts with engineering problems.

Every session follows the workflow you would use on a real project, so every line of code has an engineering purpose.

	Define the problem Which mix-design inputs drive concrete strength?	01
	Understand the data Explore, clean and check it before you trust it.	02
	Build a model Regression, trees, ensembles and neural networks.	03
	Validate it Test on data the model has never seen.	04
	Interpret it Does the model respect engineering physics?	05
	Decide responsibly Account for uncertainty before acting on a prediction.	06

EXAMPLES FROM

-  Concrete materials
-  Geotechnical
-  Transportation
-  Water & environment
-  Buildings & energy

Built for civil engineers at every stage.

Whether you are still studying or already on site, you will leave able to apply ML to your own engineering problems.



UG students
B.Tech / B.E. in civil engineering



PG students
M.Tech / M.E. in any civil specialisation



PhD scholars
and early-career researchers



Faculty members
bringing data-driven methods into teaching



Practising engineers
structural, geotech, transport, water



Complete beginners
never written a line of Python? Perfect.

 **What you need**

- ✓ Basic civil engineering concepts
- ✓ Comfort with everyday engineering calculations
- ✓ A Google account and a web browser

 **What you don't**

- ✗ ~~Prior Python or programming~~
- ✗ ~~Prior machine learning experience~~
- ✗ ~~Advanced mathematics~~
- ✗ ~~Anaconda, MATLAB or any installs~~

By the end, you'll be able to...

	Frame engineering problems for ML Turn a civil engineering question into inputs, features and a target.	01
	Wrangle real engineering data Explore, clean and prepare datasets with Python, NumPy and pandas.	02
	Predict engineering quantities Build regression models for strength, response and performance.	03
	Support engineering decisions Build classifiers and judge them with precision, recall and F1.	04
	Choose the right model Compare linear, tree-based, ensemble and neural-network approaches.	05
	Validate like an engineer Catch data leakage, overfitting and unsafe extrapolation.	06
	Explain what the model learned Use feature importance and SHAP to test physical plausibility.	07
	Quantify uncertainty Estimate probability of failure with Monte Carlo and ML surrogates.	08

Sessions 01–04: the foundations.

Python, data preparation, regression and nonlinear models, on real civil engineering data.

01

AI & ML for civil engineers: from problems to Python

 **Case study** · Concrete compressive strength

What AI, ML and deep learning really mean for engineers, then your first steps in Colab.

AI/ML concepts

Google Colab

Python

NumPy

pandas

First plots

02

From raw engineering data to an ML-ready dataset

 **Case study** · Concrete + cross-domain engineering data

Learn to question a dataset before you model it.

EDA

Data quality

Outliers

Correlations

Train/test split

Scaling

Leakage

03

Your first engineering prediction model

 **Case study** · Concrete compressive strength

Fit, score and diagnose a regression model the right way.

Linear regression

MAE · MSE · RMSE

R^2

Residuals

Cross-validation

04

Nonlinear engineering problems: trees & ensembles

 **Case study** · Concrete + geotechnical response

When straight lines fail: models that capture nonlinear behaviour.

Decision trees

Random Forest

Boosting

Model comparison



Milestone: by session 04 you will have built, validated and compared linear, tree and ensemble models on real concrete data.

Sessions 05–08: prediction to decision.

05 From prediction to engineering decision: classification

 **Case study** · Environmental / water + geotechnical classification

Move from “how much?” to “safe or unsafe?”.

Logistic regression

Decision trees

Confusion matrix

Precision · Recall · F1

06 Artificial neural networks: from neurons to prediction

 **Case study** · Concrete & engineering response prediction

Open the black box: how a network learns, and when it overfits.

ANN architecture

Activation

Loss

Epochs

Overfitting

Convergence

07 Can we trust our model? Explainability & judgement

 **RESEARCH AREA**

 **Case study** · Geotechnical research case study

Check whether a model’s reasoning makes engineering sense.

Feature importance

Permutation importance

SHAP

Extrapolation

Physical plausibility

08 From prediction to probability: uncertainty quantification

 **RESEARCH AREA**

 **Case study** · Reliability / engineering exceedance

Why a single number is not enough, and what to report instead.

Monte Carlo simulation

Surrogate models

Probability of failure

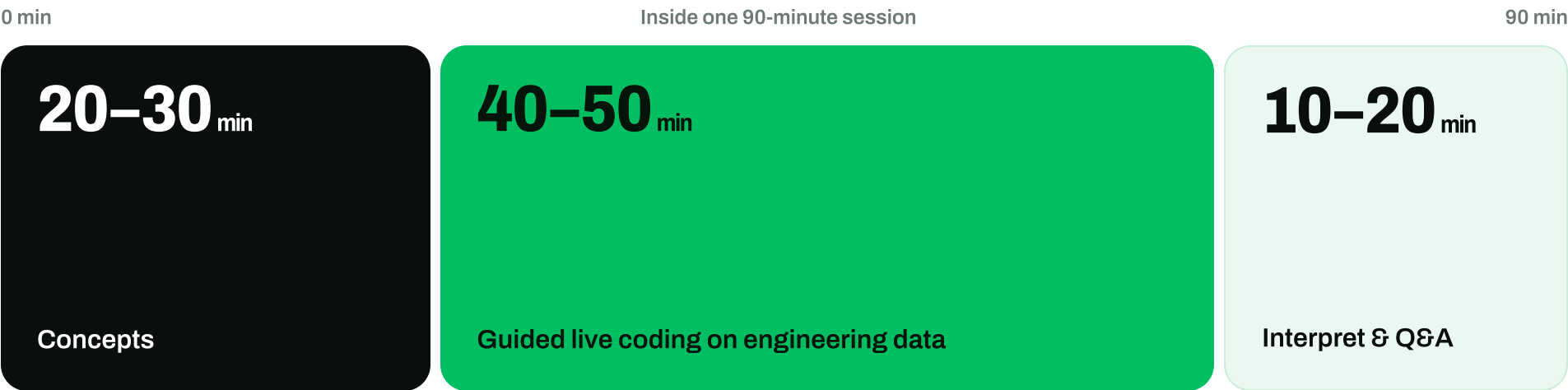
Uncertainty



Beyond a typical ML course: sessions 07 and 08 draw directly on the instructor’s research in explainable ML and reliability analysis.

Fewer slides. More code.

Every 90-minute session is live and instructor-led, and roughly half of it is spent writing code together.



Short concept explanations, then hands-on coding, then what the results mean for an engineer.

Live instructor-led coding

Code is written live with you, not pasted from slides.

Step-by-step notebooks

Every step explained in plain language inside the notebook.

Worked example, then your turn

Guided exercises follow every demonstration.

Assignments with solutions

Practise on your own, then compare with the instructor's.

Engineering discussion

Ask questions and debate what the results really mean.

Recordings for revision

Missed a session? Catch up for up to a month.

Engineering judgement is built in. Every session discusses model limitations, validation, extrapolation and physical plausibility, not just accuracy.

Real engineering data. Right in your browser.

● Google Colab · Python 3

```
[1] import pandas as pd
df = pd.read_csv("concrete_strength.csv")
df.shape

(1030, 9)

[2] from sklearn.model_selection import train_test_split
X, y = df.drop(columns="strength_mpa"), df["strength_mpa"]
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2)

[3] from sklearn.linear_model import LinearRegression
model = LinearRegression().fit(X_tr, y_tr) # your first model
```

SciPy

08 / 11

Dr. Abdul Waris

Research Associate
Universität Siegen (University of Siegen)
Former Prime Minister’s Research Fellow · Ph.D., IIT Hyderabad

A computational geomechanics researcher who uses machine learning, reliability analysis and uncertainty quantification to make engineering decisions safer. His Ph.D. developed data-driven frameworks for the reliability analysis of slopes under rainfall.

- Scientific ML
- Explainable AI
- Surrogate modelling
- Reliability analysis
- Uncertainty quantification
- Offshore geotechnics



ACADEMIC JOURNEY

2016 – 2020

B.Tech, Civil Engineering

IUST, Awantipora

Ranked 2nd in a batch of 120+

2020 – 2022

M.Tech, Geotechnical Engg.

NIT Durgapur

Institute Gold Medalist

2022 – 2026

Ph.D., Geotechnical Engg.

IIT Hyderabad

PM Research Fellow · Research Excellence Award 2026

2026 – NOW

Research Associate

Universität Siegen, Germany

Teaches & supervises Master’s students

- WORKS WITH
- Python
- MATLAB
- PLAXIS
- ABAQUS
- HYDRUS
- GIS
- AutoCAD

Taught by someone who publishes this work.

10	8	9	4
international journal papers	in Q1-ranked journals	conference papers & book chapters	countries where he has presented

PUBLISHED WITH **ASCE** **Springer** **Elsevier** **Nature Portfolio**

Selected ML publications

Explainable ensemble learning for estimating strength and stiffness of clay-bearing rocks

Scientific Reports, Nature Portfolio · 2026

Reliability analysis of unsaturated soil slopes using advanced surrogate models

Journal of Computing in Civil Engineering, ASCE · in press

Detection and prediction of slope stability using interpretable machine learning

Modeling Earth Systems and Environment, Springer · 2025

Pseudo-static slope stability analysis using explainable ML techniques

Journal of Natural Hazards, Springer · 2024

Invited lectures & talks

Machine Learning for Geotechnical Engineers: Fundamentals & Practical Applications

Oregon State University, USA · Invited talk · 2024

Five-day workshop: ML & Finite Element Methods in Geotechnical Engineering

NIT Srinagar · Resource person · 2025

Unlocking Geotechnical Insights with Machine Learning

D.Y. Patil Institute of Technology, Pune · National FDP · 2025

Conference presentations

Oslo, Norway · Barcelona, Spain · 2025

Teaching experience


Master’s teaching & supervision
Universität Siegen, Germany


Course instructor, Structural Analysis
B.Tech students, University of Kashmir


Teaching assistant, Geotechnical Engg.
IIT Hyderabad & NIT Srinagar

Everything you need, included.

No software to buy and nothing to install. Every notebook opens straight in Google Colab.



8 teaching notebooks

Structured Colab notebooks, one per session



Curated datasets

Cleaned, with data dictionaries



Exercises & assignments

Practice problems for every topic



Solution notebooks

Full instructor solutions to compare



Python & ML quick reference

A cheat sheet to keep at your desk



Session recordings

One month of access for revision

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-  8 live sessions
-  90 min each
-  4 weeks
-  Certificate
-  100% online

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Seats are limited so everyone gets real support during live coding.